

ASK 2027

Workshop on Agentic Methods for Sparse and Irregular HPC Kernels

Wednesday morning, June 2, 2027 • IPDPS 2027 • Seattle, Washington

https://ypsoh.github.io/ask-ipdps-2027/

Call for Papers

ASK 2027 invites original, unpublished work on agentic methods for high-performance computing kernels, with particular emphasis on sparse and irregular workloads. The workshop brings together researchers in sparse and combinatorial computation, compilers and performance tools, and agentic AI systems. Negative results and failure analyses are explicitly welcome.

Topics include

- Compile–run–profile–revise optimization loops
- Counter- and roofline-guided agentic search
- Sparse format and algorithm selection
- Input-dependent kernel specialization
- Sparse tensor kernels and decompositions
- Multi-agent generation, verification, tuning, and porting
- Compiler, profiler, performance-model, and sparse-DSL tool use
- Correctness and numerical verification
- Performance portability across CPUs, GPUs, and accelerators
- Benchmarks and evaluation for HPC coding agents
- Cost, energy, and human-in-the-loop tradeoffs

Dense, stencil, and FFT kernels are in scope when the primary contribution is the agentic method.

Submission tracks

Full papers: up to 10 pages for complete research results and substantial systems, methods, benchmarks, or empirical studies.

Short papers: up to 4 pages for work in progress, position papers, experience reports, negative results, and emerging ideas. Accepted short papers will also be presented at the cross-workshop poster session.

Formatting

Use the official IEEE conference proceedings template: US Letter (8.5 × 11 inches), 10-point type, single-spaced, double-column. Page limits include the body, figures, and tables; references are excluded. Appendices must fit within the applicable limit. Submit one PDF through Linklings.

ASK uses **single-anonymous review**; include author names, affiliations, and acknowledgments. Submissions must not be published or simultaneously under review at another archival venue.

Review and publication

Every paper will receive at least three reviews. Accepted full and short papers will appear in the IPDPS Workshops proceedings (IPDPSW), submitted for inclusion in IEEE Xplore. At least one author of each accepted paper must register and present in person.

Authors should report model and tool versions, sampling parameters, optimization budgets, hardware/software environments, correctness criteria, and the number of trials. Code, matrices, prompts, agent trajectories, and other artifacts are strongly encouraged.

Important dates

Submission	Jan. 29, 2027, 11:59 p.m. AoE
Notification	Feb. 22, 2027
Camera-ready	Mar. 6, 2027
Poster session	June 1, 2027, 6:00–8:00 p.m.
Workshop	June 2, 2027, morning

Organizers and contact

Ramakrishnan Kannan (ORNL), Yongseok Soh (ORNL), and Ariful Azad (Texas A&M University).
Questions: sohyp@ornl.gov. The workshop-specific Linklings URL will be posted on the workshop website.